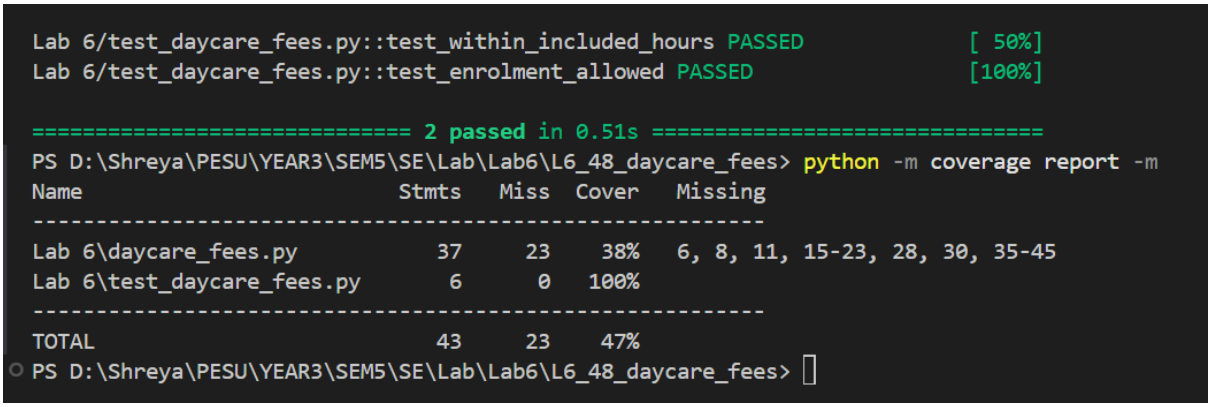
SE LAB 6

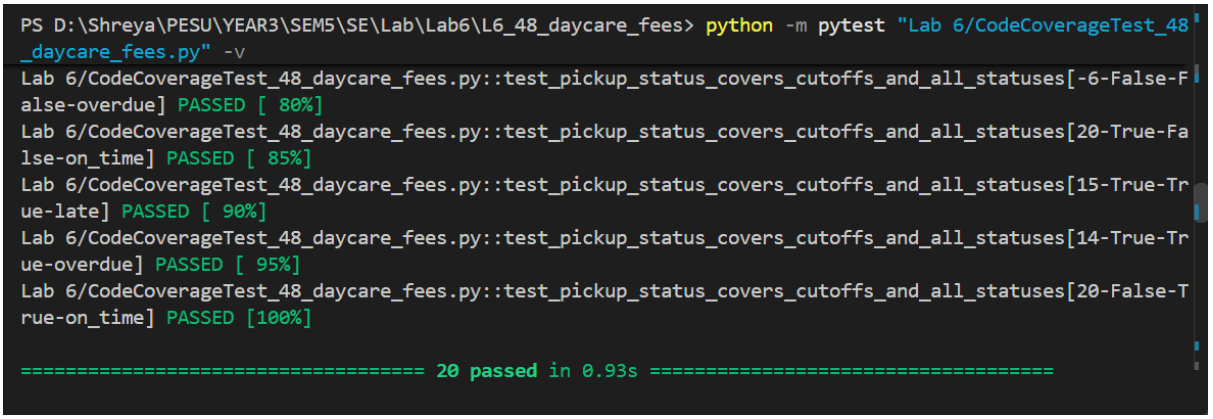
Name: Shreya Shenoy
SRN: PES2UG24CS487
Class: 5H

Date: 29-10-2026



initial code coverage report

The initial test suite consisted of two test cases, both of which passed successfully. The application achieved **38% line coverage**, with 23 out of 37 executable statements remaining untested. This indicated that additional test cases were required to cover the application's remaining functions, conditional statements and edge cases.



The dedicated code coverage test suite was executed using pytest, with all 20 test cases passing successfully.

```
========================================= 20 passed in 0.27s =========================================
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees> python -m coverage report -m
Name                                      Stmts   Miss  Cover   Missing
-----------------------------------------------------------------------
Lab 6\CodeCoverageTest_48_daycare_fees.py     26      0   100%
Lab 6\daycare_fees.py                         37      0   100%
-----------------------------------------------------------------------
TOTAL                                         63      0   100%
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees>
```

Coverage.py confirmed 100% statement coverage of the daycare fees application, with all 37 executable statements covered by the test suite

PHASE 2:Branch coverage

```
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees> python -m coverage run --branch --include="
*daycare_fees.py" -m pytest "Lab 6/BranchCoverageTest_48_daycare_fees.py" -v
se-False-overdue] PASSED [ 80%]
Lab 6/BranchCoverageTest_48_daycare_fees.py::test_pickup_status_covers_cutoff_and_status_branches[20-Tru
e-False-on_time] PASSED [ 85%]
Lab 6/BranchCoverageTest_48_daycare_fees.py::test_pickup_status_covers_cutoff_and_status_branches[15-Tru
e-True-late] PASSED [ 90%]
Lab 6/BranchCoverageTest_48_daycare_fees.py::test_pickup_status_covers_cutoff_and_status_branches[14-Tru
e-True-overdue] PASSED [ 95%]
Lab 6/BranchCoverageTest_48_daycare_fees.py::test_pickup_status_covers_cutoff_and_status_branches[20-Fal
se-True-on_time] PASSED [100%]

========================================= 21 passed in 0.42s =========================================
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees>
```

The branch coverage test suite was executed successfully, with all 21 test cases passing

```
========================================= 21 passed in 0.26s =========================================
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees> python -m coverage report -m
Name                                      Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------
Lab 6\BranchCoverageTest_48_daycare_fees.py     19      0      0      0   100%
Lab 6\daycare_fees.py                           37      0     26      0   100%
-----------------------------------------------------------------------------------------
TOTAL                                           56      0     26      0   100%
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees>
```

the daycare fees application achieved 100% statement and branch coverage, with all 26 branches fully covered and no partially covered branches.

```
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees> git log --oneline -5
2617a59 (HEAD -> main, origin/main, origin/HEAD) added branch coverage test cases
a936be4 added code coverage test cases and report
e049257 added some config files to gitignore
9c148a4 commit lab 6
PS D:\Shreya\PESU\YEAR3\SEM5\SE\Lab\Lab6\L6_48_daycare_fees>
```

according to the pdf asking for at least 2 separate commits. above attached is all my commits

summary:
the daycare fees application was tested using pytest and Coverage.py. The initial test suite achieved 38% code coverage. Additional test cases were developed to improve testing completeness, resulting in 100%

code coverage and 100% branch coverage. All test cases passed successfully, and the updated test files were committed and pushed to the individual GitHub repository.

Github repo: https://github.com/shenoyshre22/L6_48_daycare_fees